DATABASE MANAGEMENT SYSTEMS (CS301)

ER Diagram Notation Guide

Learning Objectives

Understand the ER Model, entities, attributes, relationships, cardinality constraints, participation constraints, and ER diagram notation.

Introduction to ER Model

The Entity Relationship (ER) Model is a high-level conceptual data model used to design and represent the structure of a database before implementation.

What is an Entity?

An entity is a real-world object or concept about which data is stored in a database.

Examples of Entities

- Student
- Faculty
- Course
- Department

Attributes

Attributes are properties or characteristics that describe an entity.

Types of Attributes

- Simple Attribute
- Composite Attribute
- Single-Valued Attribute
- Multi-Valued Attribute
- Derived Attribute

Relationships

A relationship represents an association between two or more entities.

Relationship Example

A Student enrolls in a Course. Here, 'enrolls' is the relationship between Student and Course.

Cardinality Constraints

- One-to-One (1:1)
- One-to-Many (1:N)
- Many-to-One (N:1)
- Many-to-Many (M:N)

Participation Constraints

Total Participation: Every entity must participate.

Partial Participation: Participation is optional.

Strong Entity

A strong entity has its own primary key and can exist independently.

Weak Entity

A weak entity depends on another entity for its existence and does not have a complete primary key of its own.

ER Diagram Symbols

Rectangle → Entity
Double Rectangle → Weak Entity
Diamond → Relationship
Oval → Attribute
Double Oval → Multi-Valued Attribute
Dashed Oval → Derived Attribute

Applications of ER Diagrams

• Database Design
• Requirement Analysis
• Information System Development
• Data Modeling

Exam Important Questions

1. Define ER Model.
2. Explain entities and attributes.
3. Differentiate strong and weak entities.
4. Explain cardinality constraints.
5. Draw an ER diagram for a college database.

Summary

ER diagrams provide a visual representation of database structures. Understanding entities, attributes, relationships, and constraints is essential for effective database design.